

Tutorial Sheet 2

Module 1

(1) Solve the following inequalities, with respect to x :

(a) $\frac{x+3}{2x-1} \leq x-1$ (b*) $x+2 < \sqrt{x+3}$ (c) $\frac{x+2}{x+1} > \frac{2x}{x-1}$

(2) For the following properties $\mathcal{P}(n)$, substitute $n = 1, 2$ and 3 and verify if they are true. Are they also true for every $n \geq 4$?

(a) $\mathcal{P}(n) : (n+1)^2 \geq n^n$ (b) $\mathcal{P}(n) : n! \geq n^n$ (c) $\mathcal{P}(n) : (\ln(1+n))^2 \geq (n-2)$

(3) Use the induction principle to show that the following properties $\mathcal{P}(n)$ are valid for every $n \in \mathbb{N}$:

(a) $\mathcal{P}(n) : 2^n \geq n$ (b) $\mathcal{P}(n) : 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$

(4) Find the natural domain D of the following functions. The natural domain is the largest possible set for which the function is defined.

(a) $y = \frac{\sqrt{2x+1}}{x^2-3}$

(b) $y = \sqrt[3]{x^2+x-2} - \sqrt{x^2-5x+4}$

(c) $y = \frac{2+x}{3^x+4^x}$

(d) $y = x^6 - 5x^4 + 36x^2 + 11$

(e) $y = \exp\left(\frac{x+3}{x-1}\right)$

(f) $y = \sqrt{\ln(x^2+1)}$

(5) Given the following functions $f(x)$ and $g(x)$, find the natural domain of the two possible compositions $f \circ g$ and $g \circ f$, i.e. $D_{f \circ g}$ and $D_{g \circ f}$.

$$(a) \quad f(x) = \frac{1}{\sqrt{x}+1}, \quad g(x) = \ln(x^2+3) \qquad (b) \quad f(x) = e^x, \quad g(x) = \frac{x^2+1}{x^2-1}$$

(6) Find the inverses of the following functions, restricted to the indicated domains.

$$(a) \quad y = \exp\left(\frac{x+5}{x-2}\right) \quad \text{for } x \in \mathbb{R} \setminus \{2\} \qquad (b) \quad y = \sqrt{x^2+1}, \quad \text{for } x \geq 0$$

$$(c) \quad y = \ln(x^2+5), \quad \text{for } x > 0 \qquad (d) \quad y = (x-2)(x+1)+1, \quad \text{for } x \leq \frac{1}{2}$$